

(b) (i) 1.
$$\text{force of neutron on atom} \left(= \frac{M(\Delta v_{\text{atom}})}{t} \right) = \frac{M(V - 0)}{t}$$

$$\text{force of atom on neutron} \left(= \frac{m(\Delta v_{\text{neutron}})}{t} \right) = \frac{m[(-v) - u]}{t}$$

force of atom on neutron = force of neutron on atom

$$\frac{m[(-v) - u]}{t} = - \frac{M(V - 0)}{t}$$

$$mu = MV - mv$$

2.
$$\text{fraction} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}mu^2} = \left(\frac{M - m}{M + m} \right)^2$$

$$= \left[\frac{(12(1.66 \times 10^{-27}) - 1.67 \times 10^{-27})}{(12(1.66 \times 10^{-27}) + 1.67 \times 10^{-27})} \right]^2$$

$$\text{fraction} = 0.715$$

3.
$$0.715^{(\text{no. of collisions})} = 10^{-6}$$

$$\text{no. of collisions} = \frac{\ln(10^{-6})}{\ln(0.715)}$$

$$\text{no. of collisions} = 41.2 \approx 42$$

4. M is much larger m

therefore v is approximately the same as u

(ii) In a fission reaction, a large amount of energy is always released.

Total kinetic energy of the fission products is always larger than the initial energy of the uranium and the neutron. Hence this collision is not elastic.

(c) (i) Bullet loses KE, block gain KE and work is done against friction.

Loss in kinetic energy of block and bullet is converted to gravitational potential energy of the block and bullet.

$$(ii) \quad \text{loss in KE} = \text{gain in GPE}$$

$$\frac{1}{2} Mv^2 - 0 = Mgh$$

$$v = \sqrt{2(9.81)(0.036)}$$

$$v = 0.84 \text{ m s}^{-1}$$

$$(iii) \quad m_{\text{bullet}} v_{\text{bullet}} = (M_{\text{block}} + m_{\text{bullet}}) v_{\text{final}}$$

$$v_{\text{final}} = \frac{1.0 + 2.6 \times 10^{-3}}{2.6 \times 10^{-3}} (0.84)$$

$$v_{\text{final}} = 324 \text{ m s}^{-1}$$

$$(iv) \quad KE_{\text{initial}} = KE_{\text{final}} + WD_{\text{against friction}}$$

$$\frac{1}{2} m_{\text{bullet}} v_{\text{ini}}^2 = \frac{1}{2} (m_{\text{bullet}} + m_{\text{block}}) v_{\text{fin}}^2 + WD$$

$$\begin{aligned} WD &= \frac{1}{2} (2.6 \times 10^{-3}) (324^2) - \frac{1}{2} (2.6 \times 10^{-3} + 1) (0.84^2) \\ &= 136.107 \text{ J} \end{aligned}$$

$$F_{\text{ave}} (1.4 \times 10^{-2}) = 136.107$$

$$F_{\text{ave}} = 9.72 \times 10^3 \text{ N}$$